Byung Hyun Lee

AI Researcher

Selective Knowledge Updates & Domain-Specialized Foundation Models

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KEY HIGHLIGHTS

RESEARCH SPECTRUM

Selective Knowledge Updates of Foundation Models

- **Expanding, removing and refining** knowledge while **minimizing forgetting**
- Both **post-training** (continual learning, unlearning, self-distillation & RL) and **training-free, closed-form updates**
- Extending them to **domain-specialized AI**

Model & Domain Coverage

- **Foundation models for gigapixel images, Multimodal LLMs**, **LLM reasoning**, **VLA models**, **Diffusion models**
- **Gigapixel whole slide image** analysis, **industrial defect segmentation**, robot grasping and other domain-specialized AI

KEY HIGHLIGHTS ACHIEVEMENTS & EXPERIENCE

8 **Top-tier AI papers**
NeurIPS · CVPR · ICCV · ICLR · AAAI

7 **First / co-first author**
among top-tier papers

4 **ICLR 2027 submissions**
3 as co-first author

7 **Industry & gov. projects**
3 as project leader

Research Recognition

- NeurIPS 2026 **Spotlight** · AAAI 2024 **Oral** · CVPR 2026 **Compute Transparency Champion Award**
- MICCAI 2025 REG **2nd place (team leader)** · CVPR 2026 **Outstanding Reviewer** · Qualcomm Fellowship **Finalist**

Projects

- **Project leader** — **SD Biosensor** · **MSIT · NIPA** (gigapixel histopathology whole slide image) · **KETI** (world VLA model)
- **Samsung DS** (defect segmentation) · **Samsung Research** (robot grasping) · **Ministry of Health and Welfare** (gigapixel whole slide image AI) · **US Air Force · IITP** (human-machine teaming)

Industry Experience

- **NAVER Cloud**, National AI Foundation Model Project — contributed to **HyperCLOVA X 8B Omni / 32B Think**

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06 Projects & Additional Research

PROFILE



Byung Hyun Lee

AI Researcher

ldlqudgus756@snu.ac.kr
https://hyun1a.github.io

EDUCATION

Ph.D., Electrical & Computer Engineering

Seoul National University | 2021.03 – 2027.02 (expected)

B.S., Electrical Engineering

UNIST | 2015.03 – 2018.08 | GPA 4.04 / 4.30

EXPERIENCE

Visiting Student Researcher, NAVER Cloud

National AI Foundation Model Project | 2025.10 – 2026.02

RESEARCH INTERESTS

Selective Knowledge Updates

Continual Learning
Concept Erasure · Unlearning
Training-free Weight Updates
Post-training · RL

Models & Domains

Multimodal LLMs · VLA
LLM Reasoning
Domain-specialized Models
Industrial Inspection
Diffusion Models

PUBLICATIONS

8
Top-tier conference
7 as first / co-first

4
Under review
ICLR 2027 · 3 co-first

4
Workshop / short
CVPR-W · ECCV-W · ICASSP

2
Journal papers
IEEE Access · PLOS One

2
Technical reports
HyperCLOVA X 8B · 32B

1
Patent
Granted in US · KR

PUBLICATIONS FIRST / CO-FIRST AUTHOR

CONFERENCE (SELECTED PAPERS)

- Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs **NeurIPS 2026 (Spotlight)***
- Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models **CVPR 2026***
- Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis **ICCV 2025**
- Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation **CVPR 2025***
- Concept Pinpoint Eraser for Text-to-Image Diffusion Models via Residual Attention Gate **ICLR 2025***
- Doubly Perturbed Task-Free Continual Learning **AAAI 2024 (Oral)**
- Online Continual Learning on Hierarchical Label Expansion **ICCV 2023***

* Equal contribution

UNDER REVIEW · PREPRINT

- HYPERION: Hyperbolic Grounding for Histopathological Slide Representations **ICLR 2027 submitted***
- Self-Distilled Reinforcement Learning for Language-Robust VLA Models **ICLR 2027 submitted***
- S3-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation **ICLR 2027 submitted***
- Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis **Preprint 2026***

JOURNAL

- Towards Accelerating Model Parallelism in Distributed Deep Learning Systems **PLOS One 2023***

SHORT PAPER · DOMESTIC

- Empirically Accelerating Scaled Gradient Projection Using DNN for Inverse Problems **ICASSP 2021**
- Selective Concept Erasing for Safe Diffusion Models (Best Poster Award) **KSPC 2024**

PUBLICATIONS CO-AUTHOR · TECHNICAL REPORTS · PATENTS

CONFERENCE · UNDER REVIEW

- All-in-one Image Restoration for Unknown Degradations Using Adaptive Discriminative Filters for Specific Degradations **CVPR 2023**
- RePath: Zero-Shot Whole-Slide Tumor Segmentation via Intra-Slide Relational Affinity **ICLR 2027 (sub.)**
- Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation **arXiv 2025**

JOURNAL

- Continual Test-Time Adaptation for Robust Remote Photoplethysmography Estimation **IEEE Access 2025**

TECHNICAL REPORT · HYPERCLOVA X TEAM

- HyperCLOVA X 8B Omni / 32B Think **2026**

WORKSHOP · DOMESTIC

- Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models **CVPR Workshop 2026**
- Expert Classifier Ensemble-based Post-processing Correction for Unbiased Scene Graph Generation **IPIU 2024**
- Mobile AI & AIM 2022 Challenge Reports (quantized super-resolution; depth estimation) **ECCV Workshop 2022**
- Uncertainty-based Dual-Domain Low-Dose X-ray CT Reconstruction **KSPC 2022**

PATENT

- All-in-one image quality improvement model **US 2026 · KR 2025 (granted)**

PROJECTS & AWARDS

PROJECTS

KETI (Project Leader) — **2026 – present**
Uncertainty-aware World VLA Model

MSIT · NIPA (Project Leader) — **2026**
Hyperbolic Pathology FM for Gigapixel WSI

SD Biosensor (Project Leader) — **2025**
Industry–academia collaboration

US Air Force · IITP — **2025 – present**
AI for Human-Machine Teaming

Samsung DS — **2024 – 2025**
Industrial Defect Segmentation

Ministry of Health and Welfare — **2023 – 2024**
AI System for Gigapixel Histopathological WSI Analysis

Samsung Research — **2022 – 2023**
Robot Grasping

AWARDS & HONORS

CVPR Compute Transparency Champion Award — **2026**
Erasing Thousands of Concepts (ETC)

CVPR Outstanding Reviewer (Top 5%) — **2026**

Qualcomm Innovation Fellowship Korea — **2025**
Best Paper Finalist

MICCAI REG Challenge — 2nd Place (Team Leader) — **2025**
Report generation from gigapixel WSIs · 26 teams

Apple Scholars in AI/ML PhD Fellowship — **2025**
Institutional Nominee

Yulchon AI Star Scholarship — **2025**

KSPC Best Poster Presentation Award — **2024**

ACADEMIC SERVICE

Reviewer: ICLR · NeurIPS · CVPR · ECCV · AAAI · IEEE TIP · PR

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WHY KNOWLEDGE UPDATES

Foundation models now power products on three fronts — and each must keep updating its knowledge after deployment.

Agentic / Physical AI

LLM agents · VLA-driven robots

Online knowledge addition

Adapt quickly to new tasks, tools, environments and instructions

Vertical AI

Domain-specialized models

Domain knowledge addition

New organs, types, processes and defect types

Visual Generative AI

Text-to-image / video generation

Harmful knowledge deletion

Remove copyrighted, private or unsafe concepts on demand

INDUSTRY

Deployed models should adapt to new environments — without full retraining

- **Online knowledge addition** agents & robots
- **Domain knowledge addition** medical & industrial AI

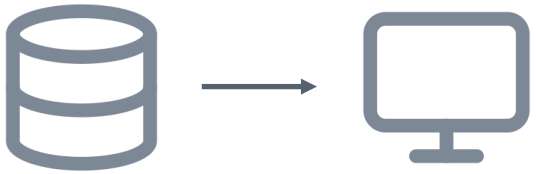
RESEARCH

- **Catastrophic forgetting** in foundation models
- **Stability–plasticity** trade-off
- **Precise, scalable unlearning**

THE CHALLENGE: Update knowledge without forgetting

1 Full retraining

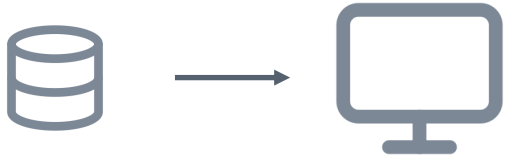
Retrain with all available data



Highly inefficient and costly

2 Naive knowledge update

Fine-tune with only the data to add or remove



Forgets existing knowledge (stability–plasticity dilemma)

3 Selective knowledge update

Update with only the target data while preserving existing knowledge



existing knowledge kept

Efficient and minimal forgetting

RESEARCH OVERVIEW

Selective Knowledge Updates & Domain-Specialized Foundation Models

Expanding, Erasing, and Refining Knowledge with Minimal Forgetting

Expanding & Refining

Continually add new knowledge

POST-TRAINING · MODEL MERGING

HLE	ICCV 2023
DPCL	AAAI 2024 Oral
DCAT	NeurIPS 2026 Spotlight
S³-OPSD	ICLR 2027 sub.
MOSAR (VLA)	ICLR 2027 sub.

Domain-Specialized AI

Gigapixel whole-slide image

POST-TRAINING FOR SPECIALIZATION

CoMEL	ICCV 2025
HYPERION	ICLR 2027 sub.
Speculative Encoding	Preprint 2026
Tree-of-Experts	REG Challenge 2025
Industrial Defect Segmentation	2025

Erasing

Remove harmful or sensitive data

CONCEPT ERASURE · UNLEARNING

CPE	ICLR 2025
GLoCE	CVPR 2025
ETC	CVPR 2026

APPLICATION

LLMs · Agentic / Physical AI | Medical AI (Pathology) | Industrial · Semiconductor AI | Visual Generative AI

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Online Continual Learning on Hierarchical Label Expansion

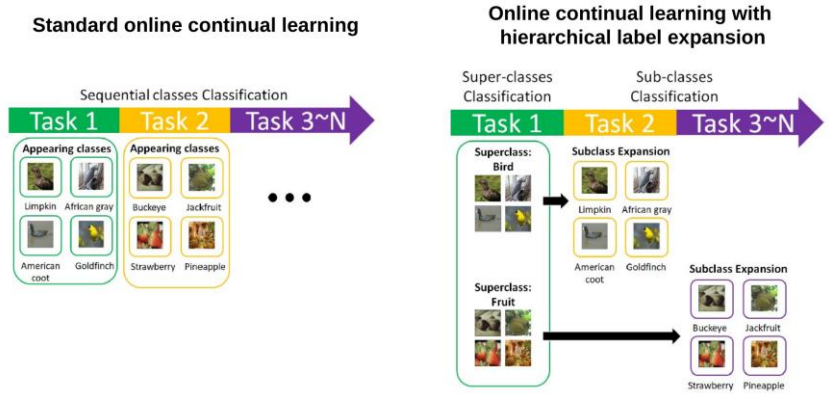
ICCV 2023 | Co-first author

EXPANDING

Keywords *online continual learning, hierarchical labels, memory management, any-time inference*

TASK OBJECTIVE

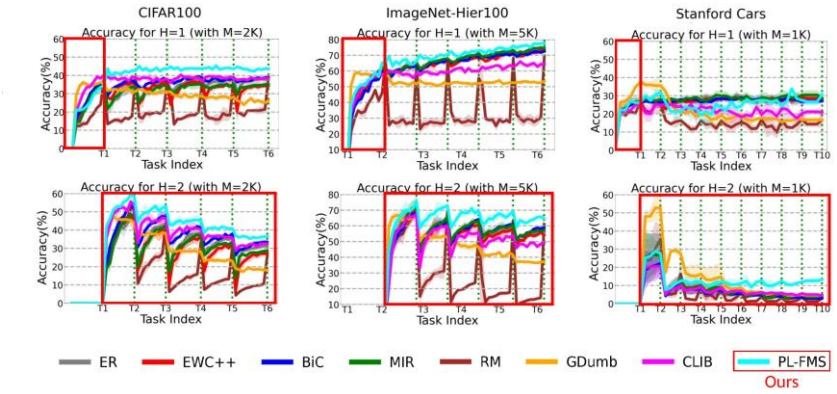
1. Define **hierarchical label expansion (HLE)** for **online continual learning**: classes are refined from coarse to fine as data streams in
2. Learn new **fine-grained classes** while retaining coarse-level knowledge



Standard online CL (left) vs. hierarchical label expansion (right)

RESEARCH SUMMARY

- New online CL setting reflecting real class hierarchies, with **single- and multi-depth** scenarios
- **Pseudo-label-based memory management** that estimates hierarchy branches and weighs samples per branch
- **Flexible Memory Sampling** that adapts the memory-to-new-data ratio over time
- **Consistently best** any-time accuracy on CIFAR100, ImageNet-Hier100 and Stanford Cars



Any-time inference (top: super-class, bottom: sub-class)

Doubly Perturbed Task-Free Continual Learning

AAAI 2024 · Oral | First author

EXPANDING

Keywords *task-free continual learning, future samples, input & weight perturbation, flat minima*

TASK OBJECTIVE

1. Go beyond preserving past samples: **account for future samples** in online continual learning
2. **Improve stability and plasticity together** with a theoretically grounded objective

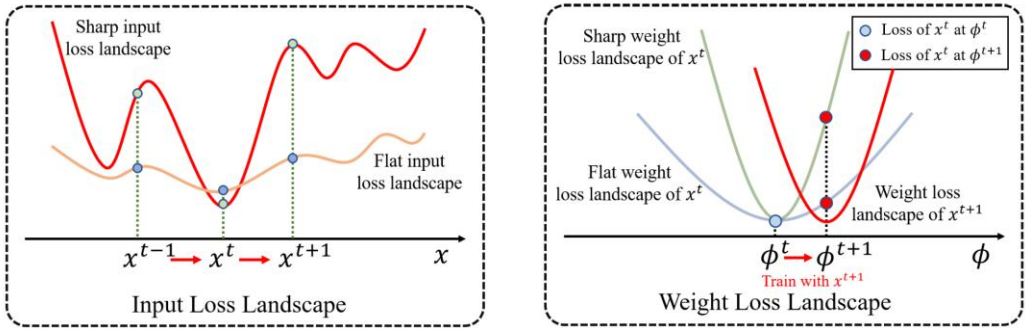
RESEARCH SUMMARY

- Proved the future-aware loss is **upper-bounded by input- and weight-perturbation terms**
- **Perturbed Function Interpolation** (feature noise + mixup) for the input term
- **Branched Stochastic Classifiers** (SWA + variational inference) for the weight term
- **Best any-time accuracy** on CIFAR100, CIFAR100-SC and ImageNet-100, with flatter loss landscapes

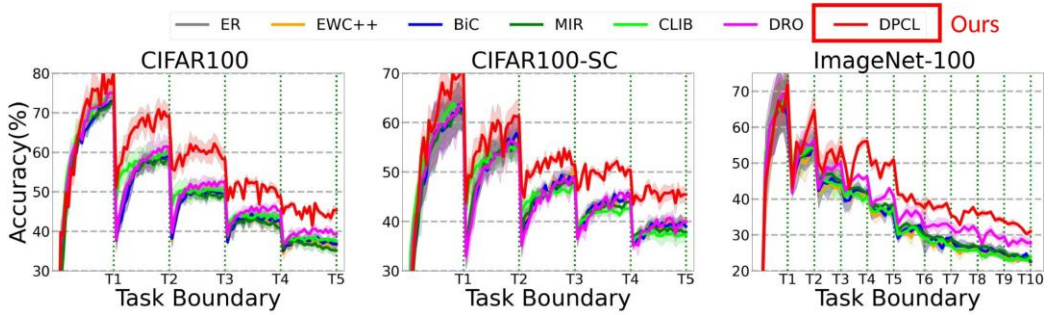
$$\mathcal{L}_t(\theta) + \lambda \max_{\|\Delta x\| \le \eta_1^t} \mathcal{L}_{t,\Delta}(\theta) + \rho \max_{\|\Delta \phi\| \le \eta_2^t} \max_{\|\Delta x\| \le \eta_1^t} \mathcal{L}_{t,\Delta}([\theta_e; \phi^t + \Delta\phi])$$

Input perturbation

Weight perturbation



Upper bound of the future-aware loss; input / weight loss landscapes



Any-time inference accuracy in class-incremental learning

Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs

NeurIPS 2026 · Spotlight | Co-first author

REFINING

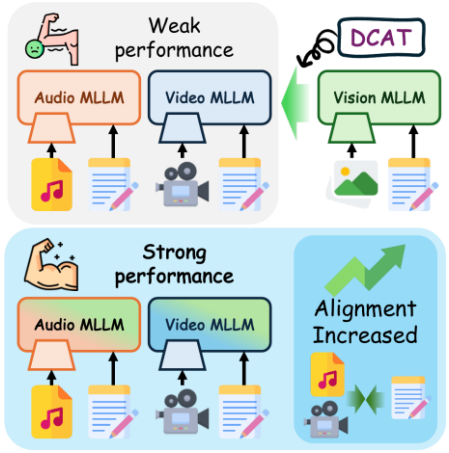
Keywords *multimodal LLM, model merging, cross-modal alignment, closed-form update*

TASK OBJECTIVE

1. Improving a weak MLLM modality (**audio, video**) usually needs large modality-specific data and compute
2. Transfer alignment from a **strong donor** (vision) to a **weak recipient** — **no fine-tuning**

RESEARCH SUMMARY

- Merging a well-aligned donor MLLM **improves the recipient on its own benchmarks**
- A **mutual-information lower bound** defines an **Alignment score** that tracks performance
- **DCAT**: a **closed-form, layer-wise weight update** from a few hundred calibration samples
- **+29.55% (audio)** and **+24.92% (video)** on average; up to **+100.92%**

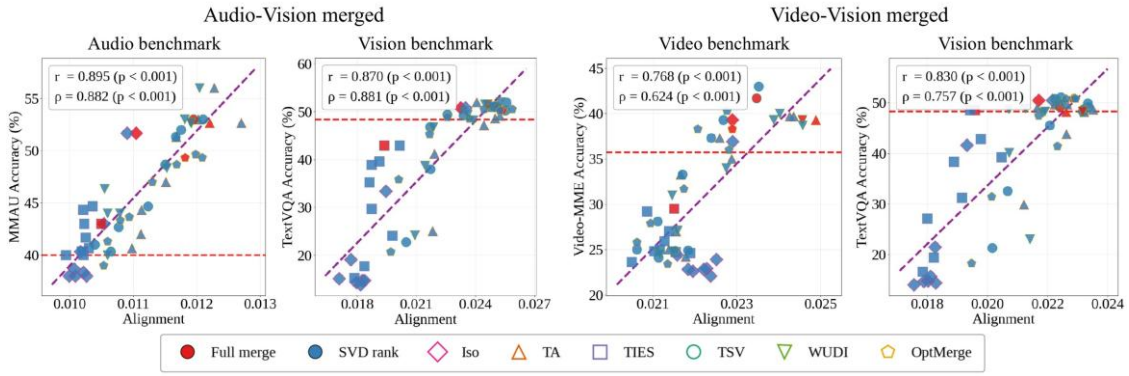


DCAT concept

+29.55% Avg. gain, audio benchmarks

+24.92% Avg. gain, video benchmarks

r = 0.895 Alignment–accuracy correlation



Alignment score vs. modality performance across merged checkpoints

S³-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation

ICLR 2027 (submitted) | Co-first author

REFINING

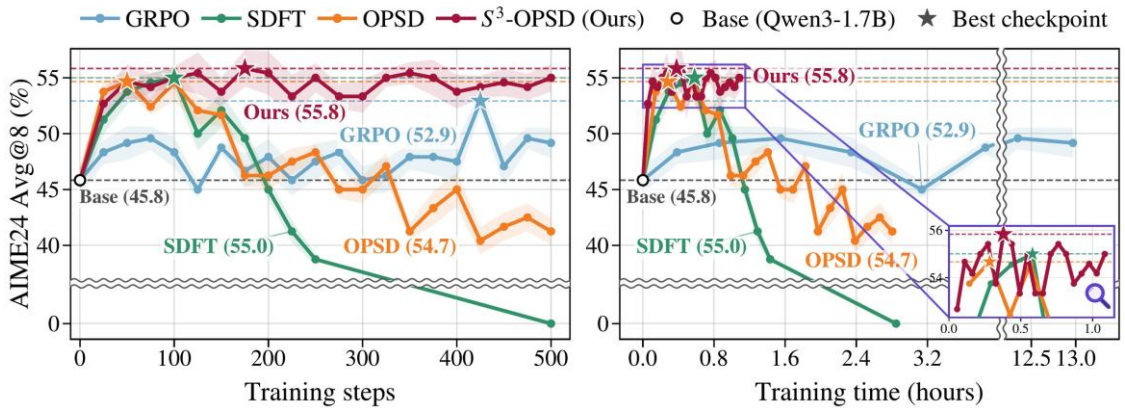
Keywords *LLM reasoning, on-policy self-distillation, sparse supervision, stability*

TASK OBJECTIVE

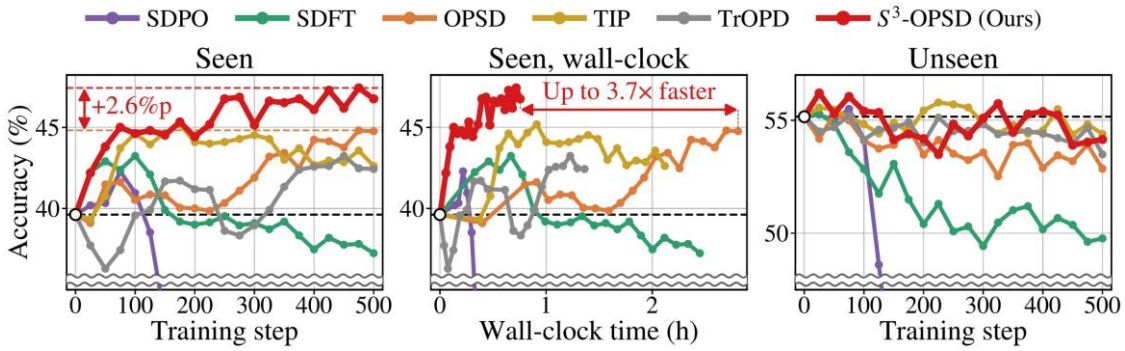
1. **On-policy self-distillation (OPSD)** improves LLM reasoning without an external teacher, but is **costly and unstable**
2. Make OPSD **sparse, stable and fast** across model families and tasks

RESEARCH SUMMARY

- **Sparse token selection** by advantage variance, via a cheap two-sided-bounded proxy
- **SGD** gives sparse updates and less drift than momentum optimizers → stability
- **Target smoothing** with the initial student as a fixed reference
- AIME24 **55.8 (+10.0 vs. base)**; up to **13× faster than GRPO**, **3× than OPSD**



AIME24 of Qwen3-1.7B vs. training steps (left) and wall-clock time (right)



Multi-task learning over seven domains

Self-Distilled Reinforcement Learning for Language-Robust VLA Models

ICLR 2027 (submitted) | Co-first author

REFINING

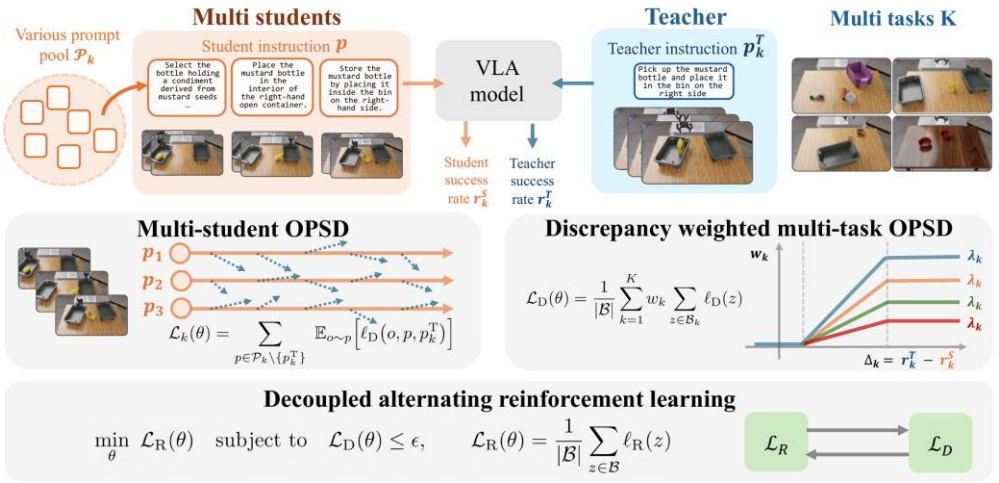
Keywords *vision-language-action, reinforcement learning, self-distillation, robotics*

TASK OBJECTIVE

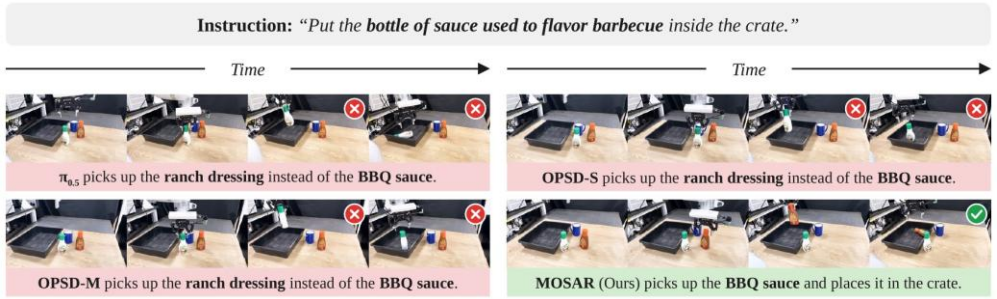
1. VLA policies are **sensitive to prompt phrasing**: paraphrases yield very different success rates
2. Close the gap, **exceed the best phrasing**, and keep unseen-task performance

RESEARCH SUMMARY

- **Multi-student OPSD**: the best prompt teaches paraphrased prompts at student-visited states
- **Alternating RL and distillation**: RL raises the ceiling, distillation spreads it
- **Discrepancy-weighted** distillation by per-task success gaps
- Unseen prompts **53.2% vs. 33.7% (PPO)**; REALM **61.0% vs. 56.8%**; **real Franka robot**



Overview of MOSAR



Real-robot evaluation: only MOSAR picks the correct object

INDUSTRY EXPERIENCE

NAVER Cloud · HyperCLOVA X

Visiting Student Researcher

NAVER Cloud | Oct 2025 – Feb 2026

Korea's National AI Foundation Model Project

~2×
faster inference, kept stable

+20%
training tokens per unit time

ROLE & CONTRIBUTIONS

- Post-training of **HyperCLOVA X**: improved **training & inference stability**
- Kept the model stable under **~2× faster inference**; built and optimized code for **deployment**
- **Dynamic batching** → **+20%** tokens trained per unit time

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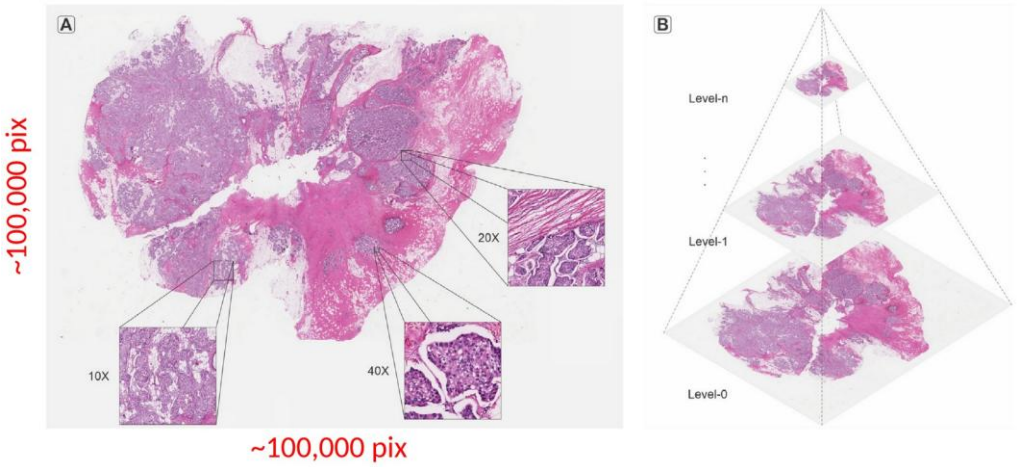
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GIGAPIXEL WHOLE SLIDE IMAGES Domain-specialized AI



Whole slide image: ~100,000 × 100,000 px

DOMAIN CHALLENGES

- Gigapixel inputs with only **slide-level labels**
- **Localizing** the evidence behind each diagnosis
- New organs and cancer types; patient-data **privacy**
- **Inference cost** of thousands of slides per day

ICCV 2025 First author

CoMEL — Continual MIL with Localization

Memory-free continual learning that keeps diagnosis and localization

ICLR 2027 sub. Co-first author

HYPERION — Hyperbolic Patch–Slide VLM

Best Dice on all five zero-shot segmentation benchmarks

Preprint 2026 Co-first author

Speculative Encoding

Up to 3× faster, 6× less memory, >99% performance retained

MICCAI 2025 2nd place · team leader

REG 2025 — Tree-of-Experts Report Generation

Structured reports from gigapixel WSIs (26 teams)

ICLR 2027 sub. Co-author

RePath — Zero-shot Tumor Segmentation

Intra-slide relational affinity for zero-shot segmentation

Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis

ICCV 2025 | First author

DOMAIN-SPECIALIZED AI & EXPANDING

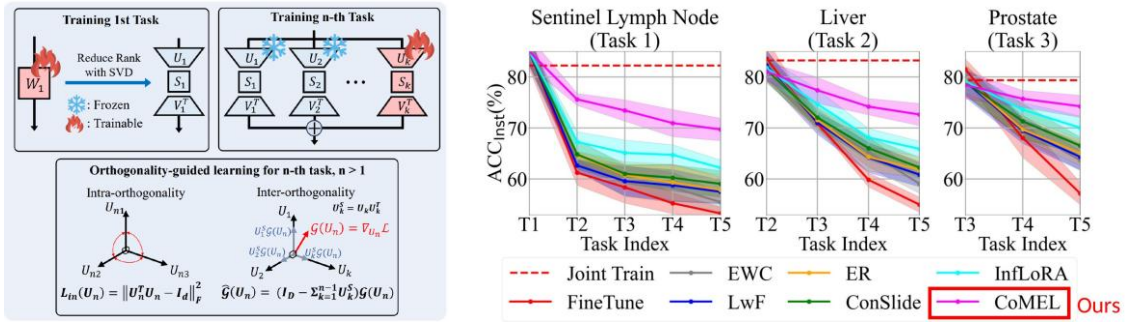
Keywords *whole slide image, multiple instance learning, memory-free CL, LoRA, localization*

TASK OBJECTIVE

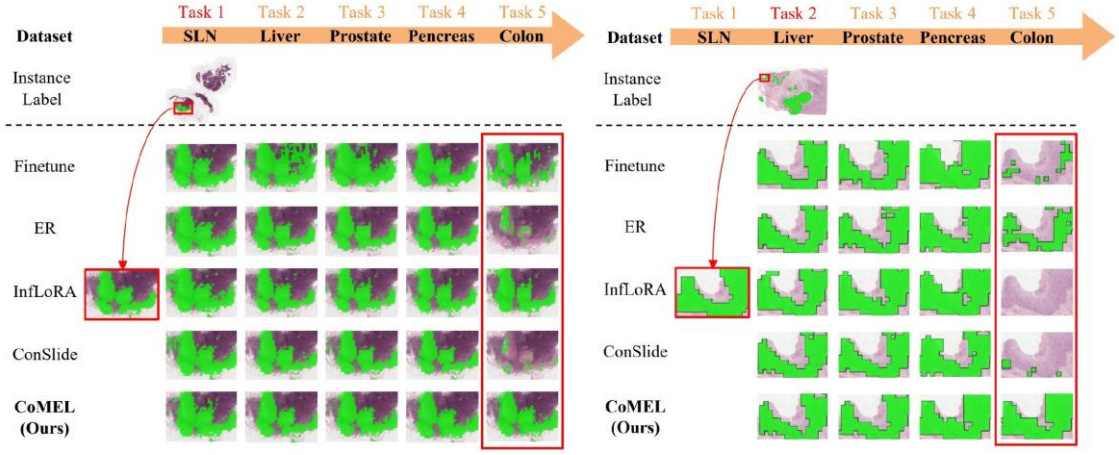
1. Continual learning for gigapixel **whole slide images** as new organs and cancer types arrive
2. **Memory-free**: keep both **diagnosis and lesion localization** without storing patient data

RESEARCH SUMMARY

- Two gaps in prior WSI continual learning: **replay-memory privacy risk** and **localization forgetting**
- **Orthogonal Weighted LoRA**: new bases orthogonal to previous tasks minimize interference
- **Bag Prototype-based Pseudo-Labeling** sharpens patch-level localization
- **Least localization forgetting** across five sequential organs (SLN → Liver → Prostate → Pancreas → Colon)



OWLoRA (left) and per-organ localization accuracy across tasks (right)



Tumor localization over the task sequence (CoMEL: ours)

HYPERION: Hyperbolic Grounding for Histopathological Slide Representations

ICLR 2027 (submitted) | Co-first author

DOMAIN-SPECIALIZED AI & REFINING

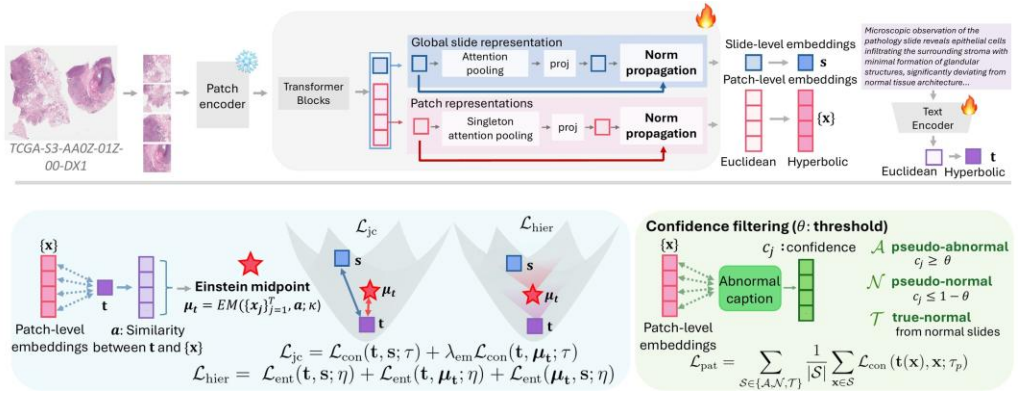
Keywords *computational pathology, hyperbolic VLM, zero-shot segmentation, report generation*

TASK OBJECTIVE

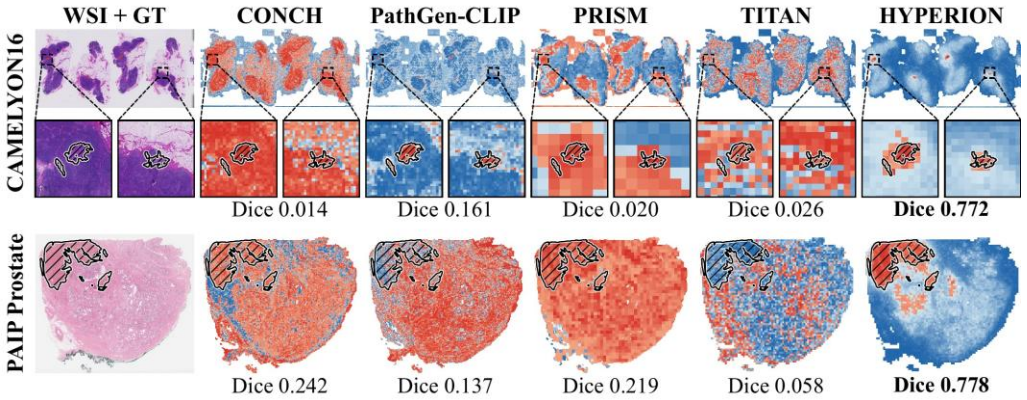
1. Euclidean slide foundation models **discard feature-norm information** that could encode hierarchy and fine detail
2. **Ground pathological concepts in patch-level evidence** for patch–slide representation

RESEARCH SUMMARY

- **Hyperbolic VLM** with contrastive alignment and **entailment cones**
- **Text-conditioned Einstein midpoints** + confidence-filtered **patch pseudo-labels**
- **Norm propagation** exposes intermediate feature norms in patch radii
- **Best Dice on all 5 benchmarks** (0.143 of baseline → 0.463); knowledge kept on **LLM tasks and prediction tasks**



Overview of HYPERION



Zero-shot tumor segmentation vs. pathology foundation models

Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis

Preprint 2026 | Co-first author

DOMAIN-SPECIALIZED AI

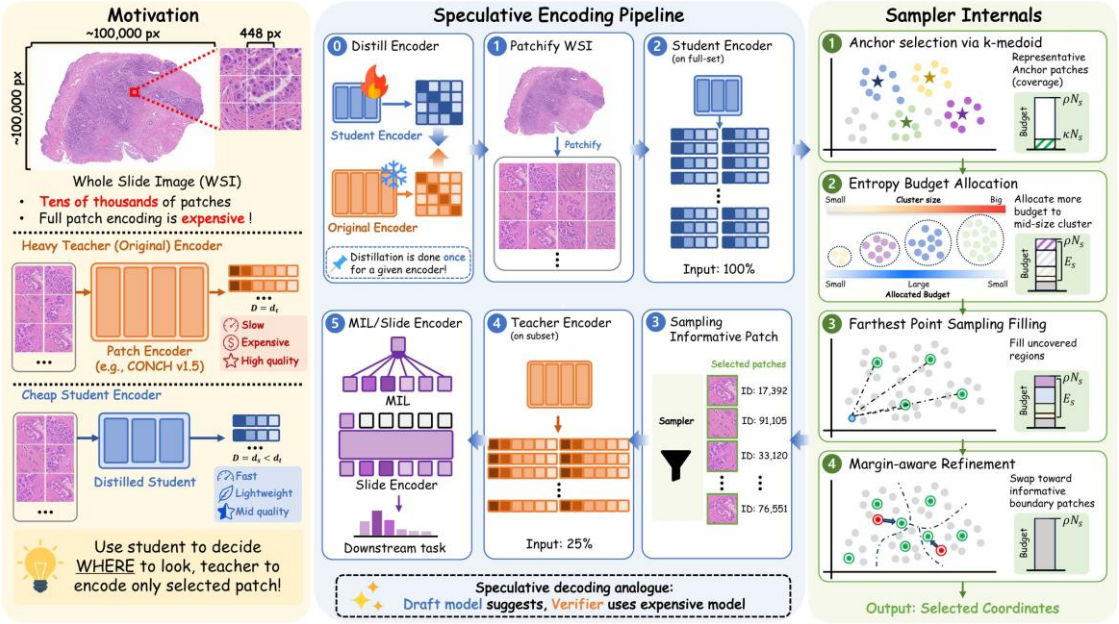
Keywords *efficient inference, whole slide image, distillation, patch sampling*

TASK OBJECTIVE

1. WSI inference is costly: **patch encoding dominates compute**, **slide aggregation drives peak memory**
2. Hospitals produce **thousands of slides per day** → fast, memory-efficient **inference** is essential

RESEARCH SUMMARY

- MIL models and slide encoders **retain performance with a sparse subset** of patches
- A **distilled light encoder drafts** features; the **full encoder runs only on selected patches**
- Sampler: k-medoid anchors, entropy budgets, farthest-point filling, margin-aware refinement
- **Up to 3× faster**, **6× lower peak memory**, **>99%** of performance (9 MIL models, 3 slide encoders)



Draft with a cheap distilled encoder, verify with the full encoder

3×
faster MIL inference
4.63 s → 1.16 s

6×
lower peak memory
TITAN: 10.4 → 1.6 GB

>99%
performance kept
9 MIL · 3 slide encoders

Tree-of-Experts for Report Generation in Histopathology

MICCAI 2025 · REG Challenge | 2nd place of the challenge · Team leader

DOMAIN-SPECIALIZED AI

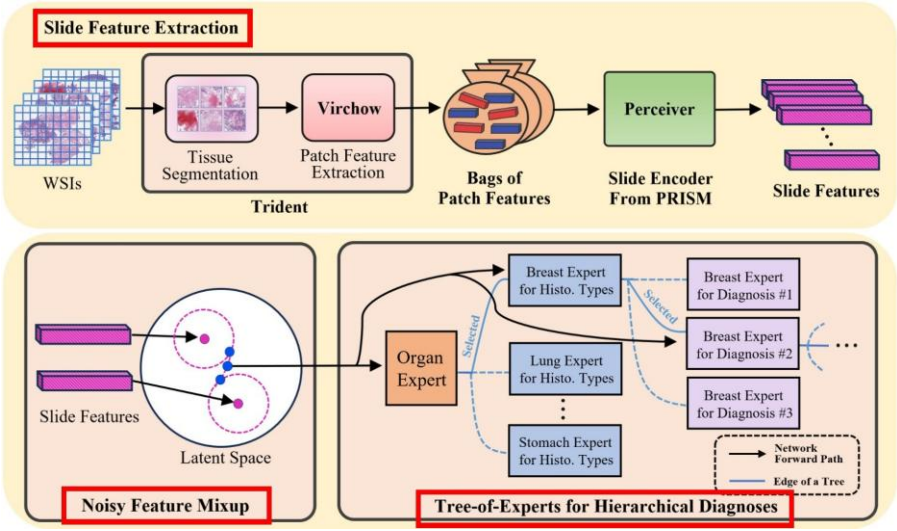
Keywords *report generation, gigapixel WSI, slide foundation model, feature augmentation*

TASK OBJECTIVE

1. Generate **structured pathology reports** (organ → procedure → histologic type → diagnoses)
2. Stay **robust across eight organs** with fewer than 10K training slides

RESEARCH SUMMARY

- Reframed generation as **hierarchical structured prediction**
- **Tree-of-Experts**: **64 lightweight MLP experts** predict report components
- **Frozen slide features** (GrandQC → Virchow → PRISM), extracted once
- **Noisy Feature Mixup** for data-scarce experts; ranking score **0.811** across 8 organs



Slide features, Noisy Feature Mixup and Tree-of-Experts

	Breast	Lung	Colon	Rectum	Stomach	Cervix	Prostate	Bladder	Avg.
Ranking	0.775	0.842	0.840	0.710	0.871	0.774	0.801	0.798	**0.811**
ROUGE	0.793	0.884	0.845	0.762	0.855	0.838	0.806	0.875	**0.832**

Validation scores per organ

DOMAIN TRANSFER

Pathology → Industrial · Semiconductor Inspection

The problem structure of gigapixel pathology could map onto high-resolution industrial and semiconductor inspection.

Gigapixel Pathology (WSI)	Industrial · Semiconductor Inspection	Related work & experience
~100,000 × 100,000 px images	High-resolution wafer, SEM, panel images	**CoMEL, HYPERION, Speculative Encoding**
Slide-level weak labels	Lot / wafer / image-level pass–fail labels	
Localizing diagnostic evidence	Defect localization and segmentation	**CoMEL, HYPERION** **Samsung DS few-shot defect segmentation**
New organs and cancer types over time	New processes, tools and defect types	**Memory-free CL (CoMEL), Generalized CL**
Patient-data privacy	Process-data security	**Continual learning without replay memory**
Scarce, costly annotation from pathologists	Scarce, costly defect masks from experts	**Training-free 1-shot segmentation, zero-shot grounding**

Training-free Few-shot Industrial Defect Segmentation

Samsung DS | Industry project · Dec 2024 – Mar 2025 DOMAIN-SPECIALIZED AI

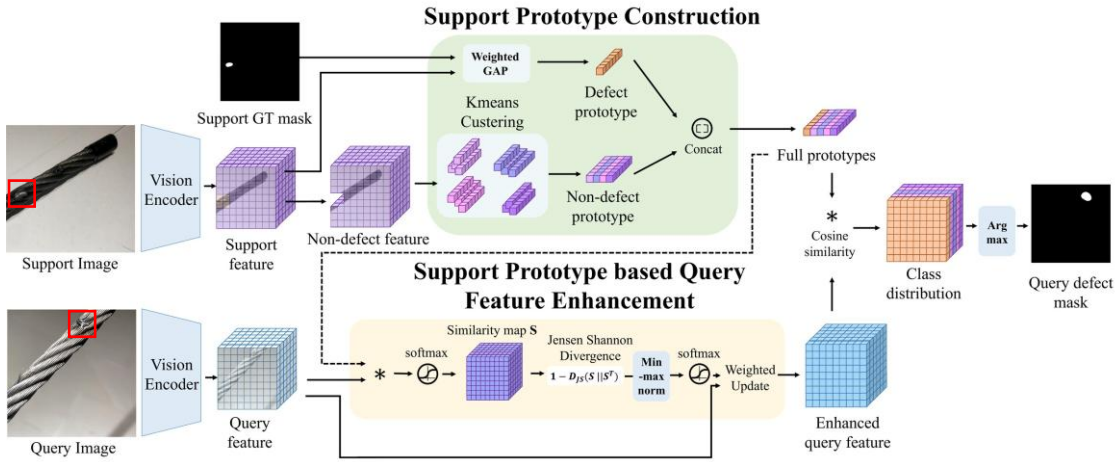
Keywords *industrial inspection, few-shot segmentation, training-free, DINOv2*

TASK OBJECTIVE

1. Segment **specific defect types** from **one annotated example** — anomaly detection cannot tell types apart
2. Avoid **training on costly masks** and handle **unseen (OOD) defects**

RESEARCH SUMMARY

- **Training-free** pipeline on a frozen **DINOv2** encoder
- **Defect + non-defect prototypes** (masked pooling, k-means) suppress false positives
- **Query enhancement** by Jensen–Shannon similarity to support prototypes
- **Best training-free mIoU**: VISION V1 **22.0** vs. 11.1 (SegGPT), on par with training-based SOTA



Support prototypes and prototype-based query enhancement

	Method	VISION V1	Defect Spectrum
Training	SOFS (SOTA)	22.5	39.1
Training-free	SegGPT	11.1	31.5
Training-free	**Ours**	**22.0**	**36.5**

1-shot defect segmentation, mIoU (%)

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Concept Pinpoint Eraser for Text-to-Image Diffusion Models via Residual Attention Gate

ICLR 2025 | Co-first author

ERASING

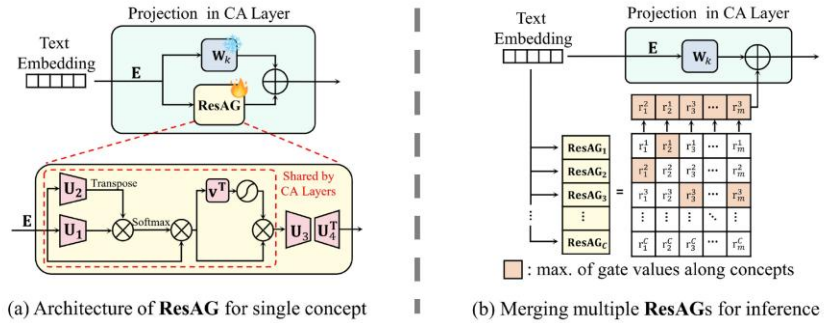
Keywords *concept erasure, text-to-image diffusion, residual attention gate, AI safety*

TASK OBJECTIVE

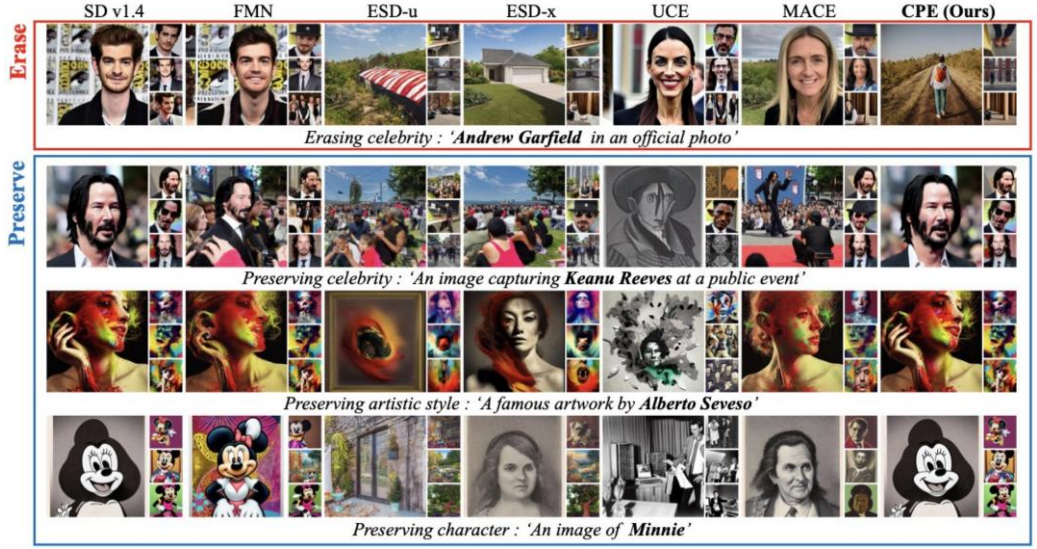
1. **Erase target concepts** (portrait rights, copyright) from text-to-image diffusion models
2. **Preserve all remaining concepts** — minimal forgetting during erasure

RESEARCH SUMMARY

- Attention-only fine-tuning cannot tighten the output-change bound
  → an **erasing–preservation dilemma**
- **Residual Attention Gate (ResAG)**: an add-on module that activates only for target concepts
- Per-concept ResAGs merge in parallel for **multi-concept erasure**
- **Near-original fidelity** on remaining concepts while erasing 50 celebrities / 100 styles



ResAG (left) and merging multiple ResAGs (right)



Erasing 50 celebrities: erased vs. preserved concepts

Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation

CVPR 2025 | Co-first author

ERASING

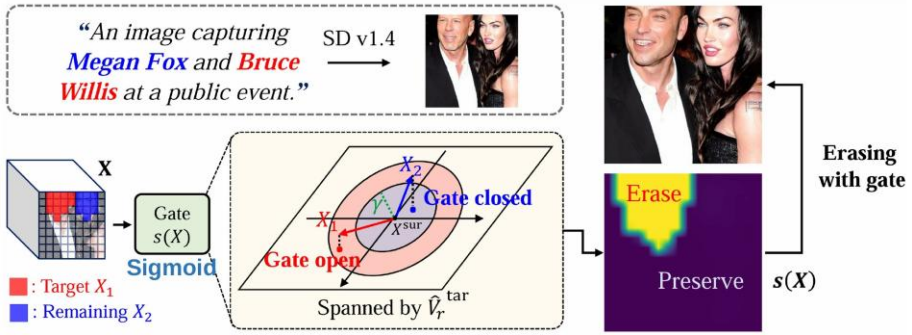
Keywords *localized concept erasure, training-free, low-rank adaptation, gating*

TASK OBJECTIVE

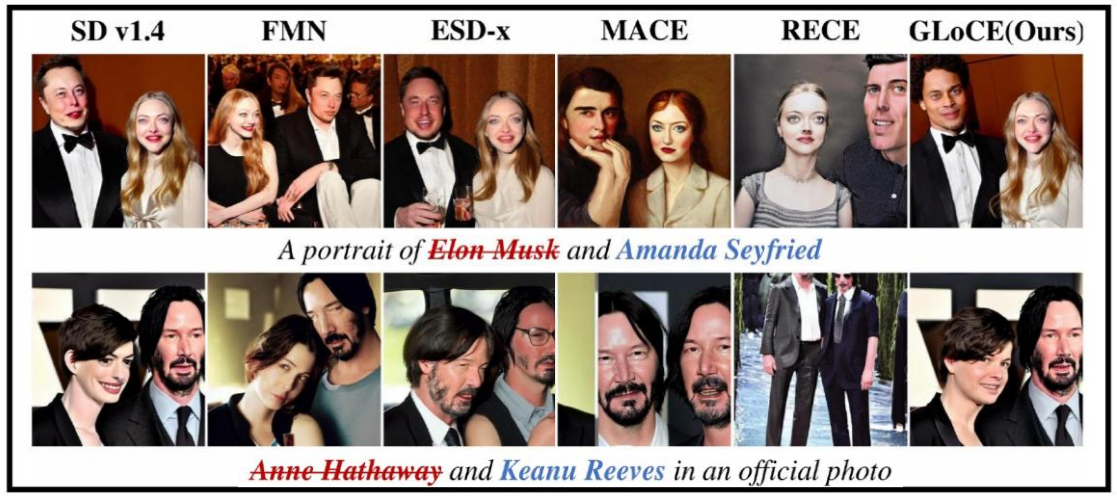
1. When several concepts share an image, **remove only the target concept's region**
2. **Keep the rest of the image intact** — without additional training

RESEARCH SUMMARY

- Concept image tokens concentrate in a **low-rank subspace**
- **Training-free low-rank update**: remove target components, project onto a mapping concept
- A **gate** that opens only on target regions → **GLoCE**, an inference-only update
- Localized erasure on **SD v1.4 / SD v3**, preserving other identities and content



Gate mechanism: erasure only on target-concept regions



Localized celebrity erasure

Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models

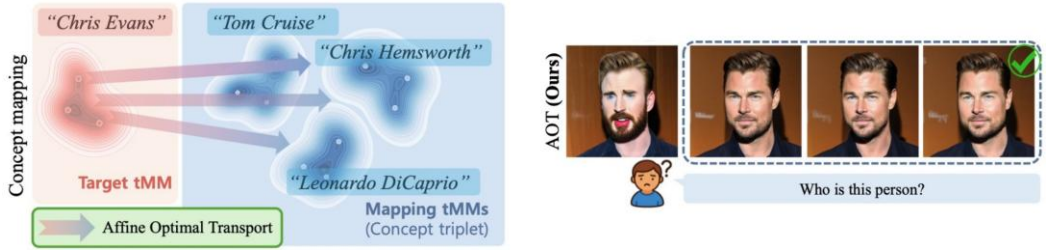
CVPR 2026 | Co-first author · Compute Transparency Champion · Popular Science Korea

ERASING

Keywords *scalable concept erasure, optimal transport, mixture-of-experts, robustness*

TASK OBJECTIVE

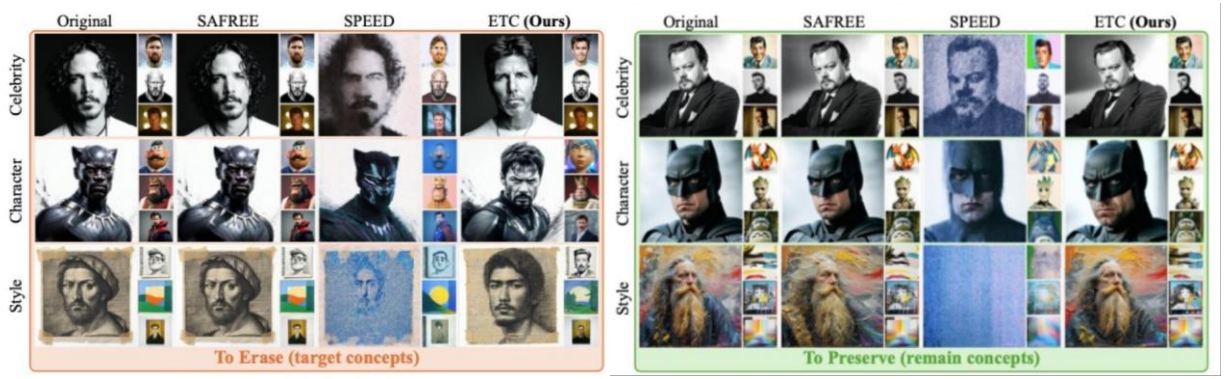
1. Scale concept erasure to **thousands of concepts** across celebrities, styles and characters
2. **Preserve quality and remaining concepts**, with **robustness** to white-box attacks



Concept mapping via affine optimal transport

RESEARCH SUMMARY

- Concept embeddings as **t-mixtures**, mapped target → mapping via **affine optimal transport**
- **Boundary anchoring** preserves neighbors without predefined anchor concepts
- **MoEraser**: one MoE module erases all targets; noise-injected training resists module removal
- Erased **2,072 concepts on SD v1.4** and **515 on SD v3.5** at once



MORE ON GENERATIVE MODELS

Unlearning · Semantic Binding · Safe Diffusion

Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models

CVPR Workshop 2026 · Co-author

- Removes outputs that **prompts cannot specify**, e.g., a particular person's face
- **Image-editing surrogates** + timestep-aware weighting + gradient surgery
- Validated on **SD3** and **DDPM-CelebA**

Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation

arXiv 2025 · Co-author

- Attributes in multi-object prompts often **bind to the wrong object**
- Analyzes the **geometry of text-token embeddings** behind attribute–object binding
- Uses it to **strengthen semantic binding** in text-to-image generation

Selective Concept Erasing for Safe Diffusion Models

KSPC 2024 · First author · Best Poster

- Erases **unsafe concepts** while keeping unrelated generations intact
- Received the **Best Poster Award**
- Early work that grew into **CPE, GLoCE and ETC**

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PROJECTS Industry · Government · Challenge

AS PROJECT LEADER / TEAM LEADER

SD Biosensor
Project Leader · 2025

Industry–Academia Collaboration

Industrial

MICCAI 2025 REG
Team Leader · 2nd / 25

Report Generation from Gigapixel WSIs

Medical

MSIT · NIPA
Project Leader · 2026

GPU Support: Pathology FM for Gigapixel WSI

Medical

KETI
Project Leader · 2026 ~

Uncertainty-aware World VLA Model

Physical · Agentic

AS PARTICIPATING RESEARCHER

Samsung DS
2024 – 2025

Industrial Defect Segmentation

Industrial · Semiconductor

Samsung Research
2022 – 2023

Robot Grasping

Physical

Korea MOHW
2023 – 2024

Gigapixel Histopathology WSI AI System

Medical

US Air Force · IITP
2025 – present

AI for Human-Machine Teaming

Agentic

IMAGE RESTORATION & EFFICIENT VISION

Low-level vision · On-device AI

CVPR 2023 · Co-author

All-in-one Image Restoration for Unknown Degradations Using Adaptive Discriminative Filters

- A single model restores images with **unknown degradation types** via degradation-specific adaptive filters
- Granted **US (2026) and KR (2025) patents**

ECCV Workshop 2022 · Challenge participant

Mobile AI & AIM 2022 Challenges — Quantized Super-Resolution on Mobile NPUs · Depth Estimation on Mobile

- Participated in **on-device** challenges: **quantized super-resolution on mobile NPUs** and **depth estimation on mobile devices**
- Co-author of the official **challenge reports**

ICASSP 2021 · First author

Empirically Accelerating Scaled Gradient Projection Using Deep Neural Network for Inverse Problems

- Accelerates iterative **inverse-problem solvers** for image processing with a learned network

KSPC 2022 · Co-author

Uncertainty-based Dual-Domain Low-Dose X-ray CT Reconstruction

- **Uncertainty-based**, **dual-domain** reconstruction for low-dose X-ray CT

CONCLUSION

01 Selective Knowledge Updates

Updating what foundation models know — **adding, removing and refining** knowledge while preserving the rest — through both **post-training** and **training-free, closed-form updates**

02 From Problem Definition to Large-scale Validation on Domain-Specialized AI

Realistic problem to **large-scale validation** (**gigapixel WSIs**, MLLMs, VLAs on a real robot, 2,000+ concepts)

03 Leadership & Industry Collaboration

Project leader for SD Biosensor, MSIT · NIPA and KETI projects; **team leader** for MICCAI 2025 REG (2nd place); contributed to **NAVER Cloud HyperCLOVA X**; projects with **Samsung DS and Samsung Research**

Thank you

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